CIRCLES

Chapter 5 — "I'm Up and Down, and Round and Round"

Ganita Manjari • Grade 9 • Part I | Notes + All 12 Theorem Proofs + All Solutions

How to use this guide: Part A explains every idea of the chapter from the very beginning — no earlier knowledge is assumed — and proves all twelve theorems in full. Part B gives worked solutions to every question in the chapter: all Think and Reflect boxes, Exercise Sets 5.1 to 5.6, and all 26 End-of-Chapter Exercises. Part C is a one-page revision sheet. Read Part A first with a pencil and paper, drawing each figure yourself as you go.

PART A — NOTES AND THEORY

5.1 WHAT IS A CIRCLE?

Everything in this chapter is drawn on a flat sheet of paper. In mathematics we call such a flat surface a plane. So all our points, lines and circles live on one plane.

Circle. A circle is the set of all points on a plane that are at the same distance from one fixed point. Think of a goat tied to a peg by a 3-metre rope: if it walks all the way round keeping the rope tight, the path it traces is a circle.

Centre. The fixed point is called the centre. (The peg.)

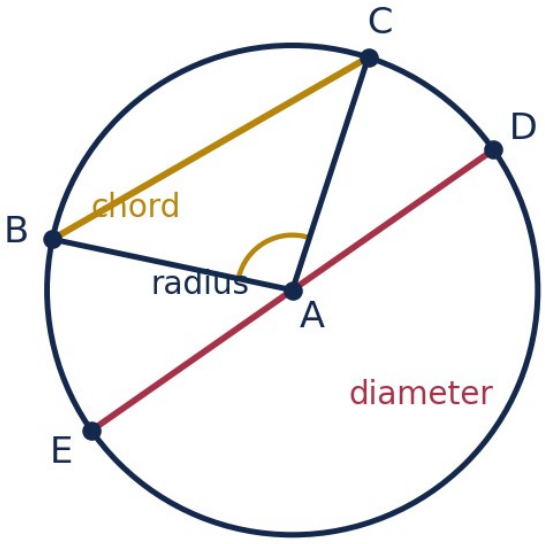
Radius. The fixed distance from the centre to any point of the circle is called the radius. (The rope.) Every radius of the same circle has the same length. We usually write it as r.

Locus. A set of points that all satisfy one particular condition is called the locus of points satisfying that condition. Using this word, a circle is the locus of points at a fixed distance from a fixed point. The word simply means "the path or the collection of all such points".

Chord. If B and C are two points on the circle, the straight line segment BC is called a chord. A chord always has both ends on the circle.

Diameter. A chord that passes through the centre is called a diameter. A diameter is the longest chord, and its length is twice the radius.

Angle subtended by a chord at the centre. Join the two ends of the chord BC to the centre A. The angle ∠BAC that is formed is called the angle subtended by the chord BC at the centre.



A is the centre; AB and AC are radii; BC is a chord; ED is a diameter

Think and Reflect (page 93): Jamuna wants to find the centre of a circular paper. Amina's suggestion: fold the paper in half so the two halves match exactly, and open it — the crease is a diameter. Fold it in half again in a different direction — the second crease is another diameter. Every diameter passes through the centre, so the point where the two creases cross is the centre.

5.2 SYMMETRIES OF A CIRCLE

Rotational symmetry. Turn a wheel by any amount and it still looks exactly the same. So a circle has complete rotational symmetry — it can be rotated about its centre through any angle at all and it looks unchanged. No other shape has this property.

Reflection symmetry. Cut out a paper circle and fold it so that the two halves overlap exactly. The crease you get is a line of reflection symmetry, and it always passes through the centre — that is, it is a diameter. Every diameter is a line of reflection symmetry, and a circle has infinitely many of them.

> **Why symmetry matters here:** Almost every theorem in this chapter is really the symmetry of the circle in disguise. If you rotate the circle, a chord moves to a new chord of exactly the same length, and its distance from the centre does not change. That single idea is behind Theorems 2, 3, 6 and 7.

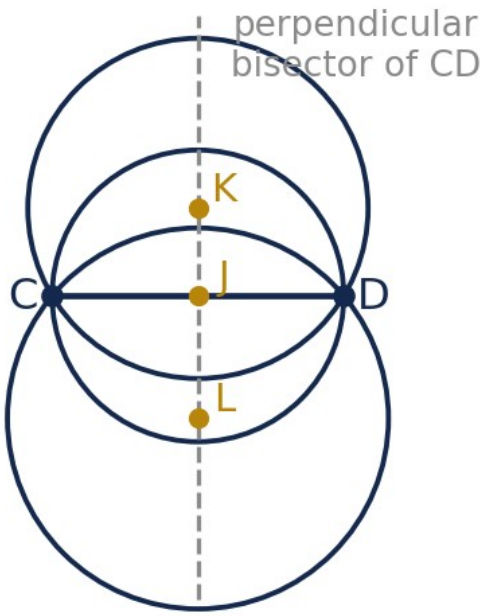
5.3 HOW MANY CIRCLES PASS THROUGH GIVEN POINTS?

Through one point. Infinitely many circles — you may choose any centre and any suitable radius.

Through two points A and B. If a circle passes through A and B, its centre O satisfies OA = OB (both are radii). So the centre must be a point equidistant from A and B. The set of all points equidistant from A and B is exactly the perpendicular bisector of AB. Therefore:

The centres of all circles through A and B lie on the perpendicular bisector of AB.

Since the perpendicular bisector contains infinitely many points, infinitely many circles pass through two given points. The smallest one has AB itself as diameter, so its radius is ½ AB. There is no largest one — as the centre moves further away along the bisector, the radius keeps growing and the circle looks flatter near A and B.



Circles through C and D: their centres J, K, L all lie on the perpendicular bisector of CD

Through three points. Now something special happens.

Theorem 1 There is exactly one circle through three non-collinear points

> **Statement:** **There is a unique circle passing through three non-collinear points.**

> **What the words mean:** Collinear means “lying on one straight line”. Non-collinear means the three points do not lie on a single straight line — so they form a proper triangle. Unique means there is one such circle and no more.

Given: Three points A, B and C that are not collinear.

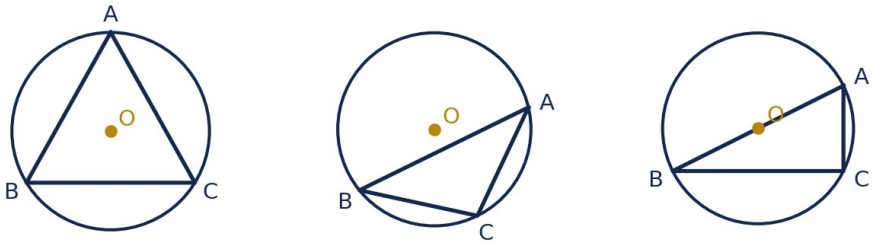
To prove: Exactly one circle passes through A, B and C.

Statement	Reason
Suppose a circle passes through A, B, C and let its centre be O.	assumption
Then OA = OB, since both are radii.	radii of the same circle
∴ O lies on the perpendicular bisector of AB.	points equidistant from A and B form this line
Also OA = OC, since both are radii.	radii of the same circle
∴ O lies on the perpendicular bisector of AC.	same reason
A, B, C are not collinear, so AB and AC are not parallel; hence their perpendicular bisectors are not parallel.	given
∴ These two perpendicular bisectors meet at exactly one point.	two non-parallel lines meet at one point only
That single point must be O — so the centre is fixed and unique.	conclusion
Taking O as centre and OA as radius, the circle drawn passes through A, B and C.	since OA = OB = OC

Hence proved.

Three non-collinear points are also the three vertices of a triangle. So we have really shown that every triangle has exactly one circle through its three vertices.

Circumcircle and circumcentre. That circle is called the circumcircle of the triangle, and its centre O is called the circumcentre. We say the circle circumscribes the triangle, and the triangle is inscribed in the circle. The circumcentre is found where the perpendicular bisectors of the sides meet.



Acute-angled: centre inside | *Obtuse-angled: centre outside* | *Right-angled: centre on hypotenuse*

Where the circumcentre lies depends on the type of triangle

- Acute-angled triangle → circumcentre lies inside the triangle.
- Obtuse-angled triangle → circumcentre lies outside the triangle.
- Right-angled triangle → circumcentre lies exactly at the midpoint of the hypotenuse.

Why no circle passes through three collinear points: If A, B, C all lie on one straight line, then the perpendicular bisector of AB and the perpendicular bisector of BC are both perpendicular to that same line, so they are parallel and never meet. There is no possible centre, so no such circle exists. This also shows a straight line can cut a circle in at most two points.

5.4 CHORDS AND THE ANGLES THEY SUBTEND

Tie a thread tightly between two points of a bicycle wheel — the thread is a chord. Join its two ends to the centre; the thread subtends an angle at the centre. Now spin the wheel. The thread turns with it, giving a new chord of exactly the same length, subtending exactly the same angle. This suggests our next theorem.

Theorem 2 Equal chords subtend equal angles at the centre

Statement: Equal chords of a circle subtend equal angles at the centre of the circle.

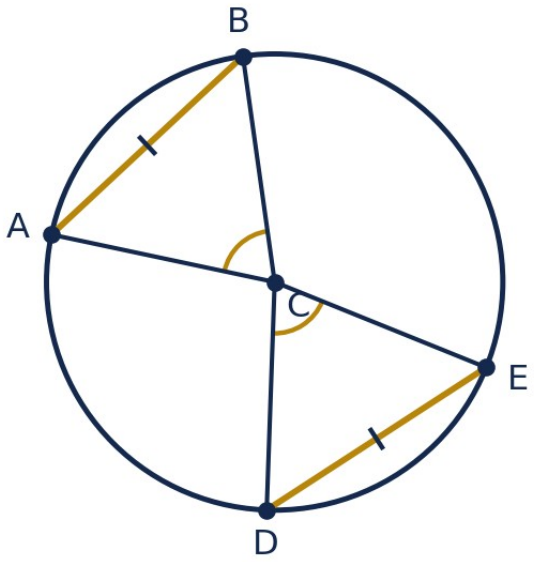
Given: A circle with centre C, and two chords AB and DE with AB = DE.

To prove: ∠ACB = ∠DCE.

Idea: The two angles sit in two different triangles, $\triangle CAB$ and $\triangle CDE$. If we can show these triangles are congruent, the angles must be equal by CPCT.

Statement	Reason
CA = CB = r	both are radii of the circle
CD = CE = r	both are radii of the circle
$\therefore$ CA = CD and CB = CE	each equals the radius r
AB = DE	given
$\therefore \triangle CAB \cong \triangle CDE$	by SSS congruence
$\therefore \angle ACB = \angle DCE$	CPCT

Hence proved.



Equal chords AB and DE subtend equal angles at the centre C

Theorem 3 Equal angles at the centre give equal chords (converse of Theorem 2)

Statement: Chords of a circle that subtend equal angles at the centre are equal.

Given: A circle with centre C, and chords AB and DE with $\angle ACB = \angle DCE$.

To prove: AB = DE.

Statement	Reason
CA = CB = r and CD = CE = r	all are radii of the circle
$\therefore$ CA = CD and CB = CE	each equals the radius r
$\angle ACB = \angle DCE$	given (this is the included angle)
$\therefore \triangle ACB \cong \triangle DCE$	by SAS congruence
$\therefore$ AB = DE	CPCT

Hence proved.

Notice the pattern: Theorem 2 uses SSS and gives equal angles. Theorem 3 uses SAS and gives equal chords. They are converses — each one is the other read backwards. Together: equal chords $\Leftrightarrow$ equal angles at the centre.

5.5 MIDPOINTS AND PERPENDICULAR BISECTORS OF CHORDS

Join the centre of a circle to the midpoint of a chord. Does anything special happen? It does — the line you draw is always perpendicular to the chord.

Theorem 4 Centre to midpoint of a chord is perpendicular to it

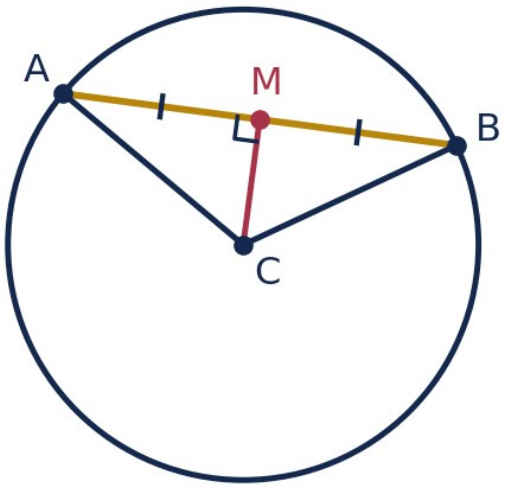
Statement: **The line joining the centre of a circle and the midpoint of a chord of the circle is perpendicular to the chord.**

Given: A circle with centre C, a chord AB, and M the midpoint of AB.

To prove: CM ⊥ AB.

Statement	Reason
CA = CB	both are radii
AM = BM	M is the midpoint of AB
CM = CM	common side
∴ △CMA ≅ △CMB	by SSS congruence
∴ ∠CMA = ∠CMB	CPCT
But ∠CMA + ∠CMB = 180°	AMB is a straight line (linear pair)
∴ each angle = 180° ÷ 2 = 90°	two equal angles adding to 180°
∴ CM ⊥ AB	an angle of 90° means perpendicular

Hence proved.



M is the midpoint of chord AB, so CM ⊥ AB

Theorem 5 Perpendicular from the centre bisects the chord (converse of Theorem 4)

Statement: **The perpendicular from the centre of a circle to a chord of the circle bisects the chord.**

Given: A circle with centre C, chord AB, and CM ⊥ AB with M on AB.

To prove: AM = BM.

Statement	Reason
∠CMA = ∠CMB = 90°	given, CM ⊥ AB
CA = CB	both are radii (these are the hypotenuses)

Statement	Reason
CM = CM	common side
∴ △CMA ≅ △CMB	by RHS congruence
∴ AM = BM	CPCT

Hence proved.

Very useful in problems: Together, Theorems 4 and 5 say that the perpendicular from the centre to a chord and the line from the centre to the chord's midpoint are the same line. So whenever you drop a perpendicular from the centre onto a chord, it automatically cuts the chord into two equal halves. Almost every numerical problem in this chapter uses this.

5.6 DISTANCE OF A CHORD FROM THE CENTRE

What "distance" means here. The distance from the centre to a chord always means the perpendicular distance — the length of the perpendicular dropped from the centre onto the chord. By Theorem 5, the foot of this perpendicular is the midpoint of the chord.

Theorem 6 Equal chords are equidistant from the centre

Statement: Chords of a circle having the same length are all at the same distance from the centre of the circle.

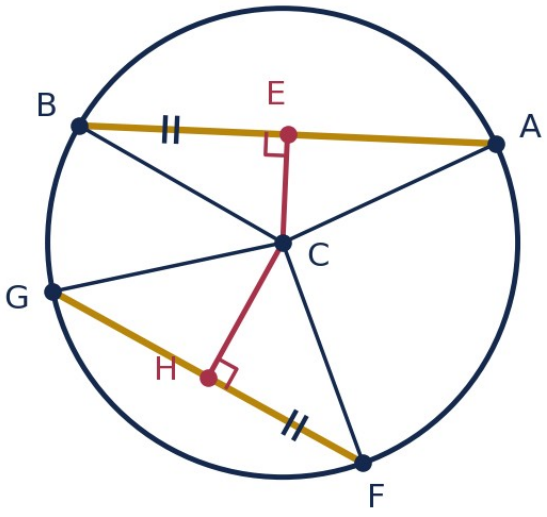
Given: Circle with centre C; chords AB and FG with AB = FG; E and H are the midpoints of AB and FG.

To prove: CE = CH.

Note: By Theorem 4, CE ⊥ AB and CH ⊥ FG, so CE and CH really are the distances of the two chords from the centre.

Statement	Reason
AE = ½ AB and FH = ½ FG	E, H are midpoints
AB = FG	given
∴ AE = FH	halves of equal lengths
∠CEA = ∠CHF = 90°	CE ⊥ AB and CH ⊥ FG (Theorem 4)
CA = CF	both are radii (the hypotenuses)
∴ △CEA ≅ △CHF	by RHS congruence
∴ CE = CH	CPCT

Hence proved.



Equal chords AB and FG are the same distance from the centre C

Second method (using Pythagoras): In right triangle CEA: $CE^2 = CA^2 - AE^2 = r^2 - (\frac{1}{2}AB)^2$. In right triangle CHF: $CH^2 = CF^2 - FH^2 = r^2 - (\frac{1}{2}FG)^2$. Since AB = FG, the two right-hand sides are equal, so $CE^2 = CH^2$, giving CE = CH.

Theorem 7 Chords equidistant from the centre are equal (converse of Theorem 6)

Statement: Chords of a circle that are equidistant from the centre have equal length.

Given: Circle with centre C; CE ⊥ AB, CH ⊥ FG, and CE = CH.

To prove: AB = FG.

Statement	Reason
∠CEA = ∠CHF = 90°	given
CA = CF	both are radii (the hypotenuses)
CE = CH	given
∴ △CEA ≅ △CHF	by RHS congruence
∴ AE = FH	CPCT
But AB = 2 AE and FG = 2 FH	the perpendicular from the centre bisects the chord (Theorem 5)
∴ AB = FG	doubling equals

Hence proved.

5.6.1 Which of two unequal chords is farther from the centre?

Theorem 8 The longer chord is nearer to the centre

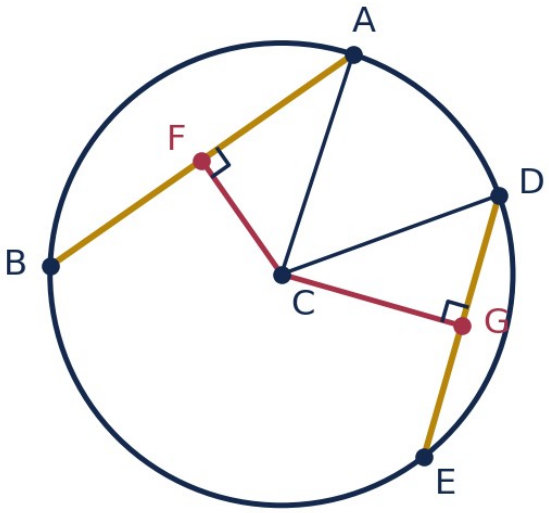
Statement: Let AB and DE be two chords of a circle with centre C, and suppose AB > DE. Then the distance from C to AB is less than the distance from C to DE.

Given: Chords AB and DE of a circle with centre C, with AB > DE. CF ⊥ AB and CG ⊥ DE.

To prove: CF < CG.

Statement	Reason
F and G are the midpoints of AB and DE.	perpendicular from the centre bisects the chord
∴ AF = ½ AB and DG = ½ DE	from the above
AB > DE, so AF > DG, and therefore $AF^2 > DG^2$	halves of a bigger length are bigger
In right △CFA: $CA^2 = CF^2 + AF^2$	Baudhāyana–Pythagoras theorem
In right △CGD: $CD^2 = CG^2 + DG^2$	Baudhāyana–Pythagoras theorem
CA = CD, both being radii, so $CF^2 + AF^2 = CG^2 + DG^2$	equating the two
∴ $CF^2 = CG^2 + DG^2 - AF^2$	rearranging
Since $AF^2 > DG^2$, the quantity $(DG^2 - AF^2)$ is negative	from step 3
∴ $CF^2 < CG^2$, and so CF < CG	lengths are positive

Hence proved.

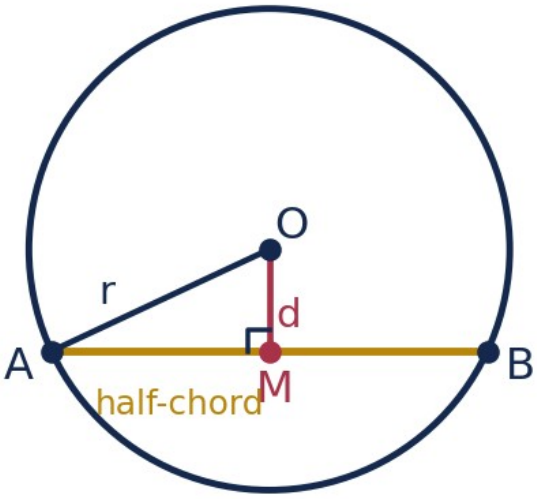


AB is the longer chord, so its distance CF is smaller than CG

Two limiting cases worth remembering: Push a chord towards the centre and it grows; the largest chord is the one through the centre — the diameter, at distance 0. Push it away and it shrinks; when its distance equals the radius it has shrunk to a single point of length 0.

The chord-length formula. From the right triangle formed by the radius, the distance d and half the chord:

$$\text{(half the chord)}^2 + d^2 = r^2 \quad \text{so} \quad \text{chord} = 2\sqrt{(r^2 - d^2)}$$

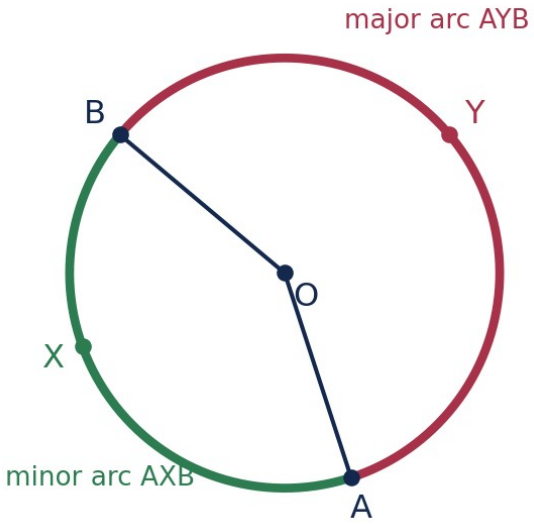


The radius, the distance d and half the chord form a right triangle

5.7 ANGLES SUBTENDED BY AN ARC

Arc. An arc is a connected piece of the circle — a portion of the curved boundary. Two points A and B on a circle cut it into two arcs. The bigger one is the major arc and the smaller one is the minor arc. To say which one we mean, we name a third point on it: arc AXB means the arc from A to B that passes through X.

Angle subtended by an arc at the centre. Sweep the radius from OA round to OB, travelling along the arc we are interested in, and measure the angle swept. If that angle is less than 180° the arc is a minor arc; if it is more than 180° it is a major arc.



A and B cut the circle into the minor arc AXB and the major arc AYB

Angle subtended by an arc at a point on the circle. Take any point D on the circle that is not on the arc AXB, and join DA and DB. The angle ∠ADB is called the angle subtended by arc AXB at the point D. The remarkable fact is that this angle is exactly the same wherever D is chosen, so long as D stays off the arc.

Theorem 9 Angle at the centre is double the angle at the circle

Statement: The angle subtended by an arc at the centre of the circle is double the angle subtended by the same arc at any point on the circle outside the arc.

Given: An arc AFB of a circle with centre C. D is a point on the circle, not on arc AFB.

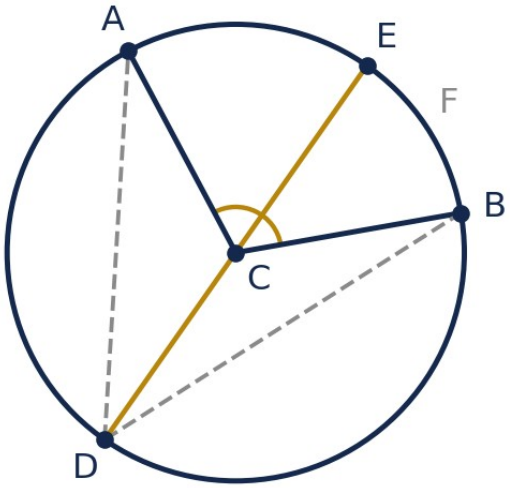
To prove: ∠BCA = 2 ∠BDA.

Construction: Join D to C and extend DC to meet the circle again at E.

Case 1: E lies on the arc AFB (see the first figure).

Statement	Reason
In △DCB, CB = CD	both are radii
∴ ∠CBD = ∠CDB	angles opposite equal sides
∠BCE is an exterior angle of △BCD	DCE is a straight line
∴ ∠BCE = ∠CBD + ∠CDB = 2∠BDC	exterior angle = sum of interior opposite angles
Similarly, in △ADC, CA = CD, so ∠CAD = ∠CDA	both are radii
∴ ∠ACE = ∠CAD + ∠CDA = 2∠CDA	exterior angle theorem again
Now ∠BCA = ∠BCE + ∠ECA	the two angles together make ∠BCA
∴ ∠BCA = 2∠BDC + 2∠CDA = 2(∠BDC + ∠CDA)	substituting
∴ ∠BCA = 2 ∠BDA	since ∠BDC + ∠CDA = ∠BDA

Hence proved.

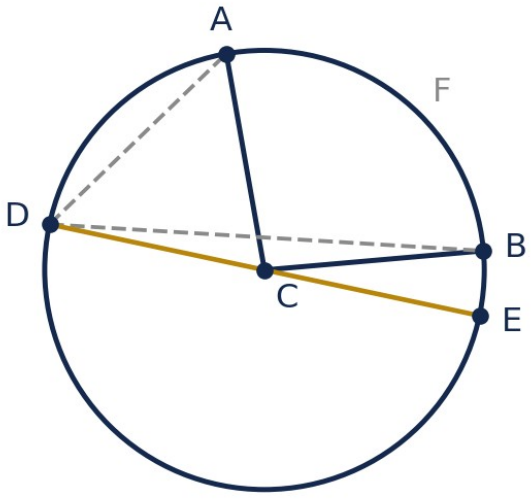


Case 1: DC extended meets the circle at E on arc AFB

Case 2: E lies outside the arc AFB. The same two exterior-angle results still hold, but now the angles are subtracted instead of added.

Statement	Reason
$\angle ACE = \angle ADC + \angle CAD = 2\angle ADC$	exterior angle of isosceles $\triangle ACD$, where CA = CD
$\angle BCE = \angle BDC + \angle CBD = 2\angle BDC$	exterior angle of isosceles $\triangle BCD$, where CB = CD
$\angle ACB = \angle ACE - \angle BCE$	from the figure
$\therefore \angle ACB = 2\angle ADC - 2\angle BDC = 2(\angle ADC - \angle BDC)$	substituting
$\therefore \angle ACB = 2\ \angle ADB$	since $\angle ADC - \angle BDC = \angle ADB$

Hence proved.



Case 2: DC extended meets the circle at E outside arc AFB

The big consequence — angles in the same segment: The angle at the centre is fixed once the arc is fixed. So every point D on the circle outside that arc gives $\angle ADB = ½$ (angle at the centre) — the same value every time. In short: angles subtended by the same arc at any points on the remaining part of the circle are all equal. This is used constantly in problems.

Corollary The angle in a semicircle is 90°

Statement: The angle subtended by a diameter at any point on the circle is 90°.

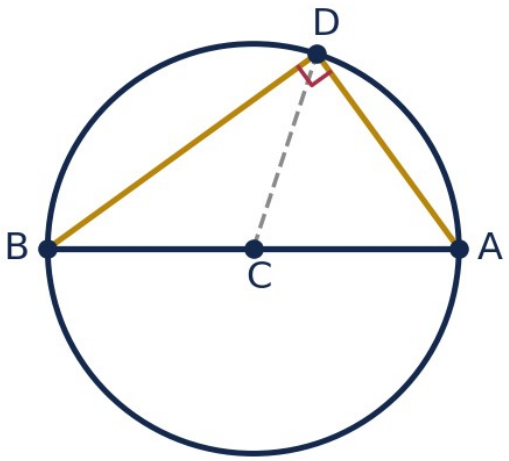
What a corollary is: A corollary is a fact that follows immediately from a result already proved — you get it almost free, with one extra line of reasoning.

Given: AB is a diameter of a circle with centre C, and D is any point on the circle.

To prove: ∠ADB = 90°.

Statement	Reason
Take the arc from A to B not containing D.	choice of arc
The angle this arc subtends at the centre is ∠ACB.	definition
A, C, B lie on one straight line, so ∠ACB = 180°.	AB is a diameter through the centre C
∠ADB = ½ ∠ACB	Theorem 9
∴ ∠ADB = ½ × 180° = 90°	substituting

Hence proved.



AB is a diameter, so the angle at D is a right angle

5.8 CONCYCLICITY AND CYCLIC QUADRILATERALS

Concyclic points. Points that all lie on one circle are called concyclic. Any three non-collinear points are automatically concyclic (Theorem 1). Four points are special — they need not lie on a circle. Theorem 10 tells us exactly when they do.

Theorem 10 Equal angles on the same side ⇒ the four points are concyclic

Statement: If a line segment AB joining two points A, B subtends equal angles at two other points C, D lying on the same side of AB, then the four points lie on a circle.

Given: Segment AB; points C and D on the same side of AB, neither on line AB; and ∠ACB = ∠ADB.

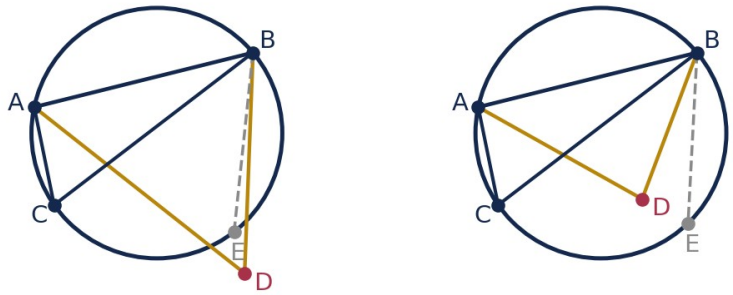
To prove: A, B, C, D are concyclic.

Method: Proof by contradiction. Draw the circle through A, B, C (possible by Theorem 1, since they are non-collinear) and show that D cannot be off this circle.

Statement	Reason
Case 1: suppose D lies outside the circle.	assumption to be disproved
Then AD cuts the circle at a point E, and E lies between A and D.	D is outside
∠AEB = ∠ACB	angles in the same segment (Theorem 9)
∠AEB is an exterior angle of △BED	E lies on AD

Statement	Reason
∴ ∠AEB > ∠ADB	exterior angle > interior opposite angle
∴ ∠ACB > ∠ADB	substituting ∠AEB = ∠ACB
But ∠ACB = ∠ADB is given, so ∠ACB > ∠ACB.	impossible
∴ D cannot lie outside the circle.	contradiction
Case 2: suppose D lies inside the circle.	assumption to be disproved
Extend AD to meet the circle at E. Now ∠ADB is the exterior angle of △BED.	D is inside
∴ ∠ADB > ∠AEB = ∠ACB, again contradicting ∠ACB = ∠ADB.	same argument reversed
∴ D lies neither outside nor inside — D must be on the circle.	only case left

Hence proved.



D outside the circle *D inside the circle*

The two impossible cases: D outside and D inside the circle

Cyclic quadrilateral. A quadrilateral (also called a 4-gon) whose four vertices are concyclic is called a cyclic quadrilateral. It is a quadrilateral that can be inscribed in a circle.

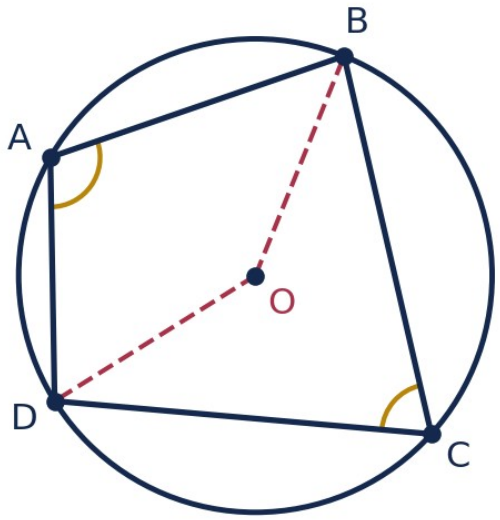
Theorem 11 Opposite angles of a cyclic quadrilateral add to 180°

Statement: The sum of two opposite angles of a cyclic quadrilateral is 180°.

Given: A cyclic quadrilateral ABCD with the circle's centre at O.
To prove: ∠BAD + ∠BCD = 180°.

Statement	Reason
A lies on the circle, outside arc BCD.	from the figure
∴ ∠BAD = ½ (reflex ∠BOD), the angle of arc BCD at the centre	Theorem 9, sweeping from OB to OD via C
C lies on the circle, outside arc BAD.	from the figure
∴ ∠BCD = ½ (∠BOD), the angle of arc BAD at the centre	Theorem 9, sweeping from OB to OD via A
Adding: ∠BAD + ∠BCD = ½ (reflex ∠BOD + ∠BOD)	adding the two results
The two angles at O together make one complete turn = 360°.	angles round a point
∴ ∠BAD + ∠BCD = ½ × 360° = 180°	substituting
Similarly ∠ABC + ∠ADC = 180°.	the same argument on the other pair

Hence proved.



The two angles at O — ordinary and reflex — together make 360°

Theorem 12 If opposite angles add to 180°, the quadrilateral is cyclic (converse of Theorem 11)

Statement: **If two opposite angles of a quadrilateral add up to 180°, then its vertices lie on a circle.**

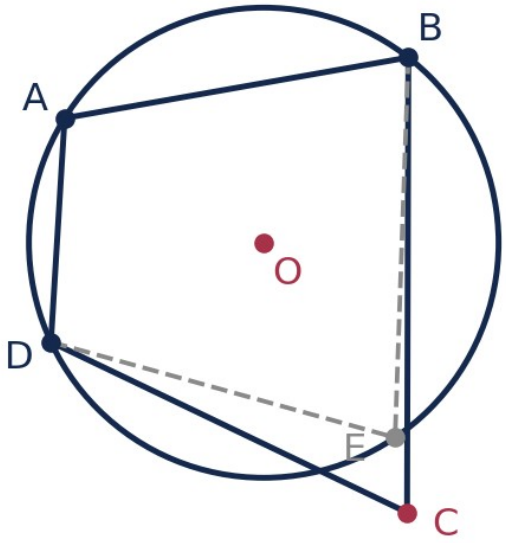
Given: Quadrilateral ABCD with ∠BAD + ∠BCD = 180°.

To prove: ABCD is cyclic.

Method: Proof by contradiction, as in Theorem 10. A, B, D are non-collinear, so draw the circle through A, B and D, then show C must lie on it. We work through the case where C is outside; the case where C is inside is exactly similar.

Statement	Reason
Suppose C lies outside the circle through A, B, D.	assumption to be disproved
Then DC cuts the circle at a point E, with E between D and C.	C is outside
ABED is a cyclic quadrilateral.	A, B, E, D all lie on the circle
∴ ∠BAD + ∠BED = 180°	Theorem 11
But ∠BAD + ∠BCD = 180° is given.	given
∴ ∠BED = ∠BCD	both make 180° with the same angle
However ∠BED is an exterior angle of △BEC	E lies on DC
∴ ∠BED > ∠BCE, that is ∠BED > ∠BCD	exterior angle > interior opposite angle
This contradicts ∠BED = ∠BCD.	impossible
∴ C is not outside; similarly C is not inside; so C lies on the circle.	conclusion

Hence proved.



C assumed outside; E is where DC meets the circle through A, B, D

PART B — COMPLETE SOLUTIONS

THINK AND REFLECT (page 94)

1. Rotational and reflection symmetries of a square, regular pentagon, regular hexagon.

   Square: rotational symmetry of order 4 — it looks the same after turns of 90°, 180°, 270° and 360°. It has 4 lines of reflection symmetry (2 diagonals and 2 lines joining midpoints of opposite sides).

   Regular pentagon: rotational symmetry of order 5 (turns of 72°, 144°, 216°, 288°, 360°) and 5 lines of reflection symmetry.

   Regular hexagon: rotational symmetry of order 6 (turns of 60°, 120°, ..., 360°) and 6 lines of reflection symmetry.

   In general a regular n-sided polygon has rotational symmetry of order n and n lines of reflection symmetry. A circle is the limiting case: infinitely many of each.

2. Longest and smallest chord in a circle of radius 5 units.

   Longest chord: the diameter, of length $2 \times 5 = 10$ units. No chord can be longer, because a chord's length is $2\sqrt{(r^2 - d^2)}$, which is greatest when $d = 0$.

   Smallest chord: there is none. As a chord is pushed further from the centre its length gets smaller and smaller, coming as close to 0 as we like, but it never reaches a smallest positive value — once the length is 0 it is only a point, not a chord.

3. Locus of points equidistant from two given points.

   It is the perpendicular bisector of the line segment joining the two points. Two things must be true for it to be the locus: (i) every point on the perpendicular bisector is equidistant from A and B, and (ii) every point equidistant from A and B lies on it. For (i): if P is on the perpendicular bisector and M is the midpoint of AB, then in $\triangle PMA$ and $\triangle PMB$ we have $AM = BM$, $\angle PMA = \angle PMB = 90°$ and PM common, so $\triangle PMA \cong \triangle PMB$ by SAS, giving $PA = PB$.

THINK AND REFLECT (page 95)

1. How many circles pass through two points on a plane?

   Infinitely many. Every point on the perpendicular bisector of AB can serve as a centre, and there are infinitely many such points.

2. Circles of all possible radii? Smallest and largest radius?

   Not all radii — only radii of at least ½ AB are possible. The smallest circle is the one with AB as a diameter, of radius ½ AB. There is no largest circle: as the centre moves further along the perpendicular bisector the radius increases without limit.

3. Moving away from AB along the perpendicular bisector — do the radii increase or decrease?

   They increase. If the centre O is at distance x from the midpoint M of AB, then the radius is $OA = \sqrt{(x^2 + (\tfrac{1}{2}AB)^2)}$, which grows as x grows.

4. Does the circle look more curved or less curved?

   Less curved. A larger radius means a flatter arc near A and B — this is why a small coin looks strongly curved while the horizon of the Earth looks almost straight.

5. How many squares with A and B on the boundary? With A and B as corners?

   On the boundary: infinitely many — A and B may be placed anywhere along the sides of squares of many different sizes and orientations.

   As corners: exactly 3. Either AB is a side of the square (giving 2 squares, one on each side of AB), or AB is a diagonal of the square (giving 1 more).

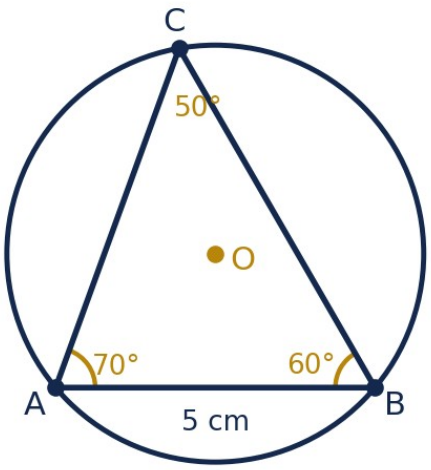
EXERCISE SET 5.1

1. $\triangle ABC$ with $AB = 5$ cm, $\angle A = 70°$, $\angle B = 60°$. Draw the circumcircle. Is the centre inside or outside?

   $\angle C = 180° - (70° + 60°) = 50°$. All three angles (70°, 60°, 50°) are less than 90°, so the triangle is acute-angled.

   Answer: the circumcentre lies inside the triangle.

Construction: draw AB = 5 cm; at A draw a 70° ray and at B a 60° ray; they meet at C. Draw the perpendicular bisectors of any two sides; they meet at O. With centre O and radius OA, draw the circle.

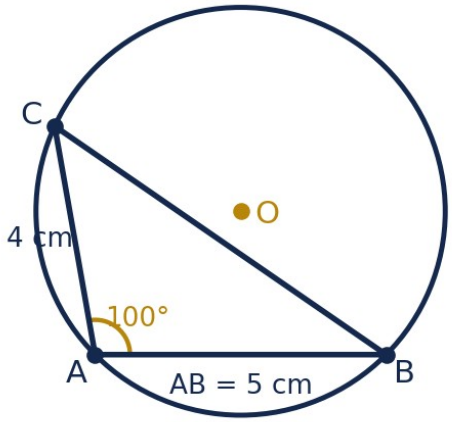


All angles are acute, so the circumcentre O lies inside the triangle

2. △ABC with AB = 5 cm, ∠A = 100°, AC = 4 cm. Draw the circumcircle. Is the centre inside or outside?

   ∠A = 100°, which is greater than 90°, so the triangle is obtuse-angled.

   Answer: the circumcentre lies outside the triangle — on the far side of BC, the side opposite the obtuse angle.



∠A = 100° is obtuse, so O falls outside the triangle

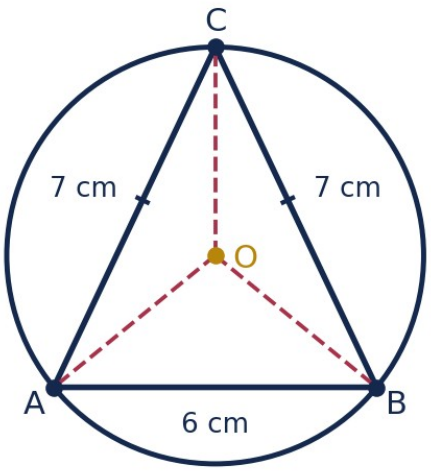
3. △ABC with AB = 6 cm, BC = 7 cm, CA = 7 cm. Find the circumcentre O and measure OA, OB, OC.

   Key point: OA = OB = OC always, because all three are radii of the circumcircle. So the three measurements must come out equal.

   Checking by calculation: the triangle is isosceles with CA = CB = 7 and base AB = 6. Its height from C is $\sqrt{(7^2 - 3^2)} = \sqrt{40} = 2\sqrt{10}$ cm, so its area = ½ × 6 × 2√10 = 6√10 cm².

   Using R = (product of the three sides) ÷ (4 × area): R = (6 × 7 × 7) ÷ (4 × 6√10) = 294 ÷ 24√10 ≈ 3.87 cm.

   Answer: OA = OB = OC ≈ 3.9 cm. The triangle is acute, so O lies inside it.

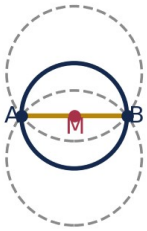


$OA = OB = OC \approx 3.9$ cm

The three distances from O to the vertices are all equal — they are radii

4. What is the least possible radius of a circle through two points A and B?

   ½ AB. The smallest circle through A and B is the one having AB as its diameter; any other circle through A and B has AB as an ordinary chord, and a chord is always shorter than the diameter, forcing a bigger radius.



smallest circle: AB is a diameter, r = ½ AB;
the dashed circles are other circles through A and B

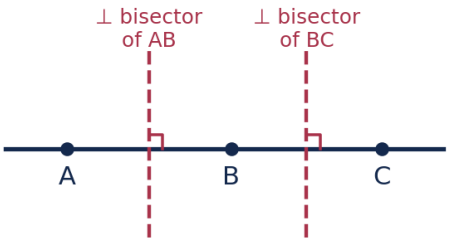
Think, Draw and Infer (page 98)

1. A, B, C collinear — is there a point P with PA = PB = PC?

   No such point exists. If PA = PB then P lies on the perpendicular bisector of AB; if PB = PC then P lies on the perpendicular bisector of BC. But A, B, C are on one straight line, so both of these bisectors are perpendicular to that same line — which makes them parallel to each other. Parallel lines never meet, so there is no point P on both.

   Therefore: no circle can pass through three collinear points, since such a circle would need a centre equidistant from all three.

   Can a line cut a circle in three points? No. A line meets a circle in at most two points — if it met it in three, those three collinear points would lie on a circle, which we have just shown to be impossible.



both are ⊥ to the same line ⇒ they are parallel ⇒ they never meet
⇒ no point is equidistant from A, B and C

2. Can other triangles congruent to △ABC share the same circumcircle?

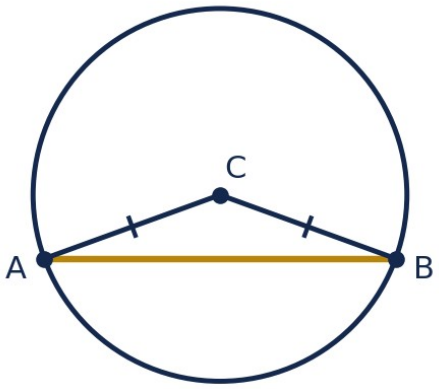
   Yes — infinitely many. Rotate △ABC about the centre O through any angle: the circle maps onto itself, and the rotated triangle is congruent to the original and still has all three vertices on the same circle. Reflecting the triangle in any diameter does the same.

EXERCISE SET 5.2

1. Show that the triangle formed by a chord and the centre of a circle is isosceles.

   Let AB be a chord and C the centre. Then CA and CB are both radii of the circle, so CA = CB. A triangle with two equal sides is isosceles, so $\triangle CAB$ is isosceles with base AB.

   Extra consequence: the base angles are equal, $\angle CAB = \angle CBA$.



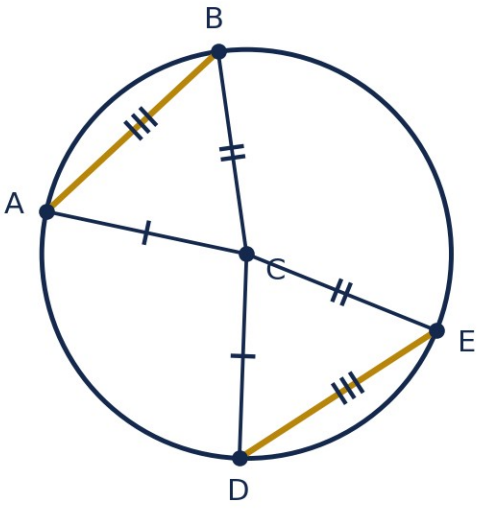
CA = CB = r ⇒ $\triangle CAB$ is isosceles

2. Show that two such isosceles triangles with equal base lengths are congruent.

   Let the two chords be AB and DE with AB = DE, and let C be the centre.

Statement	Reason
CA = CD	both are radii of the circle
CB = CE	both are radii of the circle
AB = DE	given (equal bases)
$\therefore \triangle CAB \cong \triangle CDE$	by SSS congruence

   This is exactly the proof of Theorem 2 — which is why equal chords subtend equal angles at the centre.



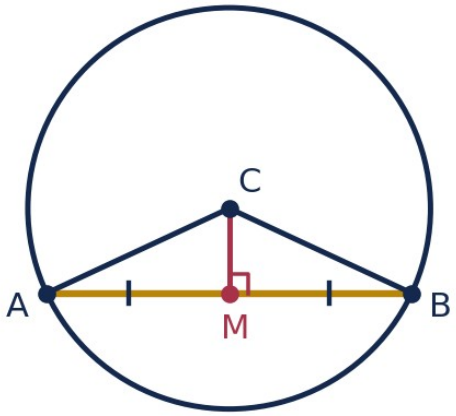
Equal chords AB and DE give congruent triangles by SSS

EXERCISE SET 5.3

1. Why is the converse of Theorem 4 true — why does the perpendicular from the centre bisect the chord?

   This is Theorem 5. Given $\angle CMA = \angle CMB = 90°$:

Statement	Reason
∠CMA = ∠CMB = 90°	given
CA = CB	both are radii — these are the hypotenuses
CM = CM	common side
∴ △CMA ≅ △CMB	by RHS congruence
∴ AM = BM, so M is the midpoint of AB	CPCT



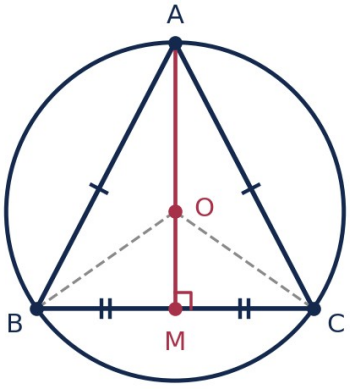
CM ⊥ AB is given; the RHS congruence then forces AM = BM

2. Isosceles △ABC inscribed in a circle with AB = AC. Show the altitude from A to BC passes through the centre.

   Since AB = AC, the point A is equidistant from B and C, so A lies on the perpendicular bisector of BC.

   The centre O is equidistant from B and C as well (OB = OC, both radii), so O also lies on the perpendicular bisector of BC.

   In an isosceles triangle the altitude from the apex A to the base BC is itself the perpendicular bisector of BC. Since O lies on that line, the altitude from A passes through the centre O.



the altitude AM passes through the centre O

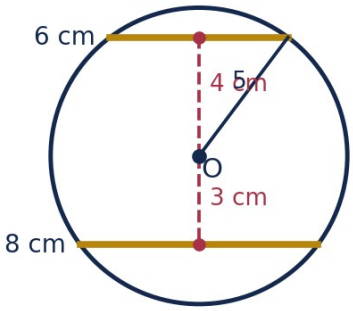
3. Two parallel chords of lengths 6 cm and 8 cm lie on opposite sides of the centre. Radius = 5 cm. Find the distance between their midpoints.

   For the 6 cm chord: half of it is 3 cm, so its distance from the centre is $\sqrt{(5^2 - 3^2)} = \sqrt{16} = 4$ cm.

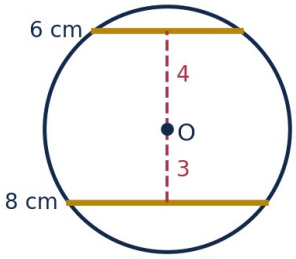
   For the 8 cm chord: half of it is 4 cm, so its distance from the centre is $\sqrt{(5^2 - 4^2)} = \sqrt{9} = 3$ cm.

   Since the chords are on opposite sides of the centre, the distances add: 4 + 3 = 7 cm.

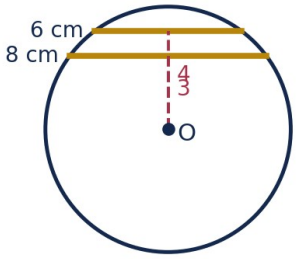
   Answer: 7 cm.



distance between midpoints = 4 + 3 = 7 cm



Opposite sides: 4 + 3 = 7 cm



Same side: 4 − 3 = 1 cm

EXERCISE SET 5.4

1. Use the Baudhāyana–Pythagoras theorem to show why Theorem 6 must be true.

Let a chord of length L be at distance d from the centre of a circle of radius r. The perpendicular from the centre bisects the chord, so it forms a right triangle with legs d and L/2 and hypotenuse r.

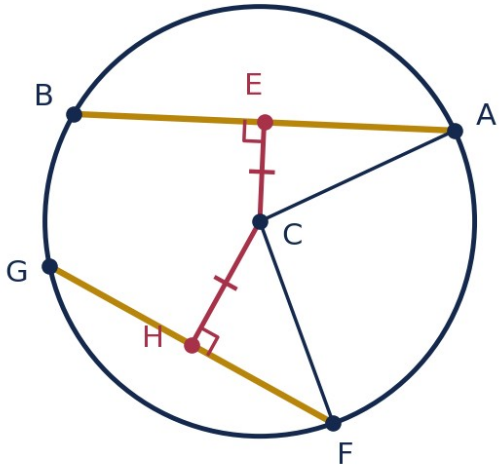
Therefore $d^2 + (L/2)^2 = r^2$, which gives $d = \sqrt{(r^2 - L^2/4)}$.

The right-hand side depends only on r and L. So two chords of the same length L in the same circle must have exactly the same distance d from the centre — which is Theorem 6.

2. In Fig. 5.15, CE ⊥ AB, CH ⊥ GF and CE = CH. Show that AB = GF.

Statement	Reason
∠CEA = ∠CHF = 90°	given, CE ⊥ AB and CH ⊥ GF
CA = CF	both are radii — the hypotenuses
CE = CH	given
∴ △CEA ≅ △CHF	by RHS congruence
∴ AE = FH	CPCT
AB = 2 AE and GF = 2 FH	perpendicular from the centre bisects the chord
∴ AB = GF	doubling equals

This is Theorem 7 — chords equidistant from the centre are equal.



CE = CH is given ⇒ AB = GF

3. Solve the previous question using the Baudhāyana–Pythagoras theorem.

In right triangle CEA: $AE^2 = CA^2 - CE^2$. In right triangle CHF: $FH^2 = CF^2 - CH^2$.

Now CA = CF (radii) and CE = CH (given), so the two right-hand sides are equal. Hence $AE^2 = FH^2$, giving AE = FH.

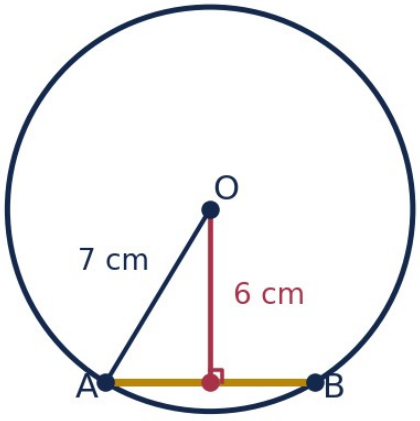
Doubling: AB = 2AE = 2FH = GF.

EXERCISE SET 5.5

1. Radius 7 cm, perpendicular distance 6 cm. Find the chord length.

Half the chord = $\sqrt{(r^2 - d^2)} = \sqrt{(7^2 - 6^2)} = \sqrt{(49 - 36)} = \sqrt{13}$.

Chord = $2\sqrt{13} \approx 7.2$ cm.



chord AB = $2\sqrt{13} \approx 7.2$ cm

2. Explain why the chord length is $2\sqrt{(r^2 - d^2)}$.

Let the chord be AB, the centre O, and OM the perpendicular from O to AB, so OM = d.

By Theorem 5, M is the midpoint of AB, so AM = ½ AB.

△OMA is right-angled at M, with hypotenuse OA = r. By the Baudhāyana–Pythagoras theorem:

$AM^2 = OA^2 - OM^2 = r^2 - d^2$, so $AM = \sqrt{(r^2 - d^2)}$.

Therefore $AB = 2\,AM = 2\sqrt{(r^2 - d^2)}$.

***3.** If the distance of chord AB from the centre is twice that of chord CD, can we conclude CD = 2 AB?

No, we cannot. The relationship between a chord's length and its distance from the centre is not a simple doubling — it involves a square root, so it is not proportional.

Counter-example: take r = 5 cm, with CD at distance 1.5 cm and AB at distance 3 cm (twice as far).

$CD = 2\sqrt{(25 - 2.25)} = 2\sqrt{22.75} \approx 9.54$ cm, $AB = 2\sqrt{(25 - 9)} = 2\sqrt{16} = 8$ cm.

Here CD ÷ AB ≈ 1.19, which is nowhere near 2.

What we can safely conclude: CD > AB — the chord nearer the centre is the longer one (Theorem 8).

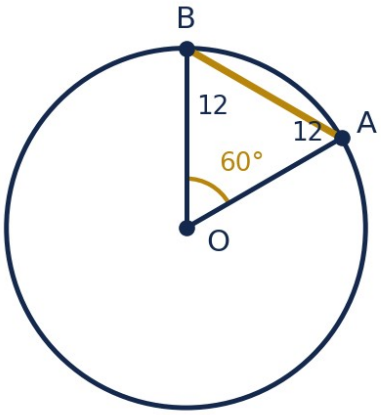
EXERCISE SET 5.6

1. Central angle ∠AOB = 60°, radius 12 cm. Find the chord AB.

In △OAB, OA = OB = 12 cm (radii), so it is isosceles and its base angles are equal.

The base angles are (180° − 60°) ÷ 2 = 60° each. So all three angles are 60° and the triangle is equilateral.

Therefore AB = OA = 12 cm.



△OAB is equilateral ⇒ AB = 12 cm

2. A and B are two points on a circle with centre O.

(i) No. All points on the same side of AB lie on the same arc, and by Theorem 9 every one of them gives the same angle. So ∠AXB and ∠AYB cannot differ.

(ii) Yes, in general. If X and Y were on opposite sides of AB then AXBY would be a cyclic quadrilateral and ∠AXB + ∠AYB = 180°, so equal angles would force each to be 90°. Apart from that one special case (when AB is a diameter), equal angles do mean X and Y are on the same side.

(iii) Yes. If ∠AXB = ∠AYB with X and Y on the same side of AB, then by Theorem 10 the four points A, B, X, Y are concyclic. Since only one circle passes through A, B and X (Theorem 1), that circle must also pass through Y.

3. Find x in Fig. 5.26, where the angle at D is 100°.

ABCD is a cyclic quadrilateral, and ∠D and ∠B are opposite angles.

By Theorem 11: ∠B + ∠D = 180°, so x + 100° = 180°.

Answer: x = 80°.

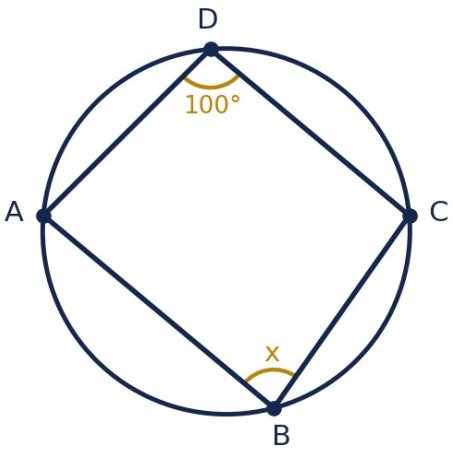


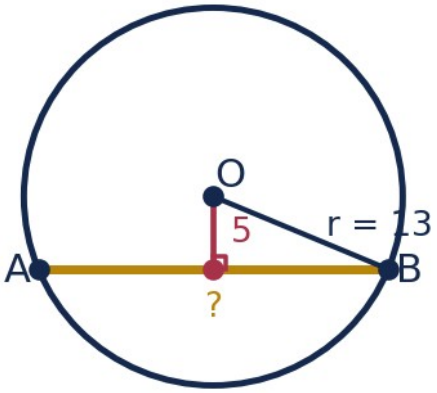
Fig. 5.26 — ∠D and ∠B are opposite angles of a cyclic quadrilateral

END-OF-CHAPTER EXERCISES (Questions 1 – 26)

1. Chord 5 cm from the centre; radius 13 cm. Find the chord length.

Half the chord = $\sqrt{(13^2 - 5^2)} = \sqrt{(169 - 25)} = \sqrt{144} = 12$ cm.

Chord = $2 \times 12 = 24$ cm.



chord = 2 × 12 = 24 cm

2. An arc subtends 70° at the centre. What angle does it subtend at a point on the circle?

By Theorem 9, the angle at the centre is double the angle at a point on the remaining part of the circle.

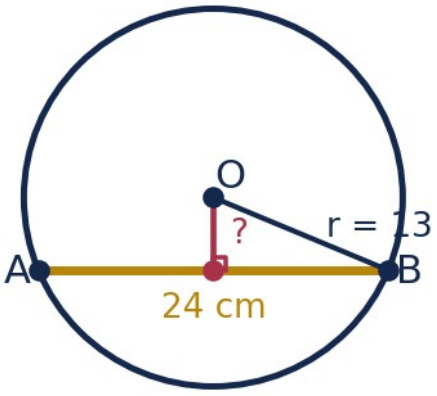
Answer: $70° \div 2 = 35°$.

3. Diameter 26 cm, chord 24 cm. Find the distance of the chord from the centre.

Radius r = 26 ÷ 2 = 13 cm. Half the chord = 24 ÷ 2 = 12 cm.

$d = \sqrt{(r^2 - (\text{half chord})^2)} = \sqrt{(169 - 144)} = \sqrt{25}$.

Answer: 5 cm.



distance = 5 cm

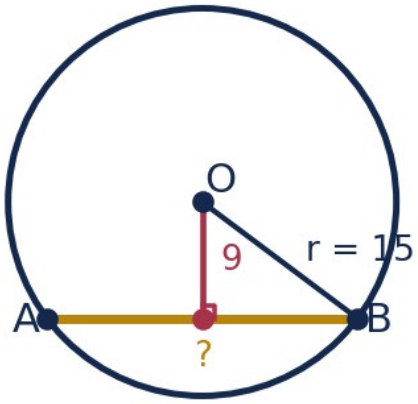
4. Radius 15 cm, chord at distance 9 cm. Find the chord length.

Half the chord = $\sqrt{(15^2 - 9^2)} = \sqrt{(225 - 81)} = \sqrt{144} = 12$ cm.

Chord = $2 \times 12 = 24$ cm.

Half the chord = $\sqrt{(15^2 - 9^2)} = \sqrt{(225 - 81)} = \sqrt{144} = 12$ cm.

Chord = $2 \times 12 = 24$ cm.



chord = 2 × 12 = 24 cm

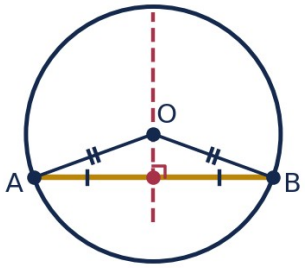
5. Prove that the perpendicular bisector of a chord passes through the centre.

   Let AB be a chord of a circle with centre O.

   OA = OB, since both are radii. So O is equidistant from A and B.

   But the set of all points equidistant from A and B is precisely the perpendicular bisector of AB.

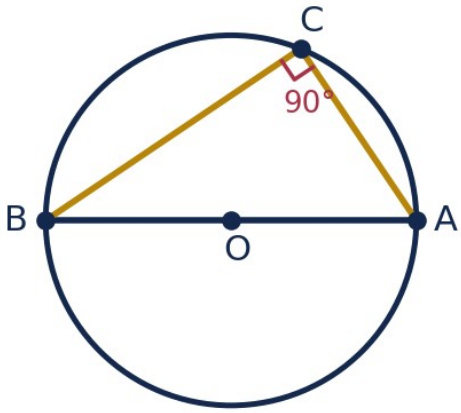
   Therefore O lies on the perpendicular bisector of AB — that is, the perpendicular bisector of the chord passes through the centre.



OA = OB ⇒ O lies on the ⊥ bisector of AB

6. AB is a diameter and C is on the circumference. Find ∠ACB.

   ∠ACB = 90°. Reason: the arc from A to B not containing C is a semicircle, and it subtends the straight angle ∠AOB = 180° at the centre. By Theorem 9, the angle at C is half of this, i.e. 90°. This is the “angle in a semicircle” corollary.



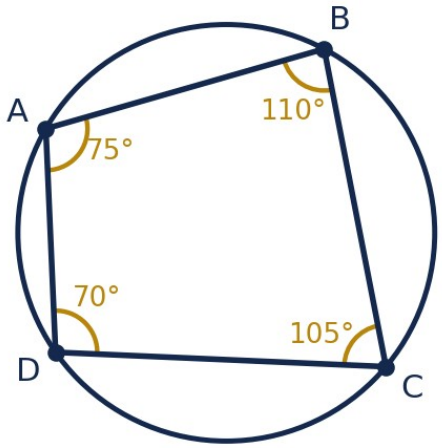
AB is a diameter, so ∠ACB = 90°

7. Cyclic quadrilateral ABCD. If ∠A = 75°, find ∠C. If ∠B = 110°, find ∠D.

Opposite angles of a cyclic quadrilateral are supplementary (Theorem 11).

∠C = 180° − 75° = 105°.

∠D = 180° − 110° = 70°.



opposite angles add to 180°

Opposite angles: 75° + 105° = 180° and 110° + 70° = 180°

8. PQRS is cyclic with ∠P = (2x + 10)° and ∠R = (3x − 20)°. Find x, ∠P and ∠R.

∠P and ∠R are opposite angles, so they add to 180°:

(2x + 10) + (3x − 20) = 180

5x − 10 = 180 ⟹ 5x = 190 ⟹ x = 38

∠P = 2(38) + 10 = 86°.

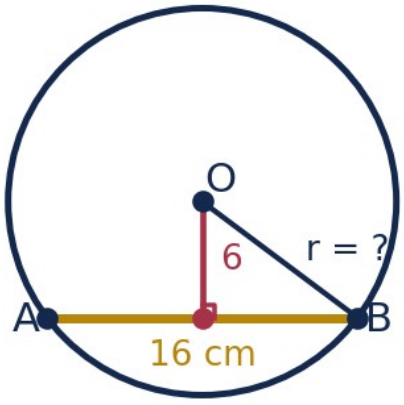
∠R = 3(38) − 20 = 94°. Check: 86° + 94° = 180° ✓

9. A chord of length 16 cm is 6 cm from the centre. Find the radius.

Half the chord = 8 cm. Using $r^2 = d^2 + (\text{half chord})^2$:

$r^2 = 6^2 + 8^2 = 36 + 64 = 100$

Answer: r = 10 cm.



$r = \sqrt{(6^2 + 8^2)} = 10$ cm

10. A cyclic quadrilateral has sides 5, 5, 12, 12 units. Find its area.

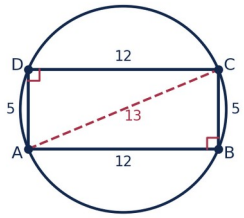
For a cyclic quadrilateral we use Brahmagupta's formula: Area = √((s−a)(s−b)(s−c)(s−d)), where s is half the perimeter.

s = (5 + 5 + 12 + 12) ÷ 2 = 34 ÷ 2 = 17

Area = √((17−5)(17−5)(17−12)(17−12)) = √(12 × 12 × 5 × 5) = 12 × 5

Answer: 60 square units.

A neat check: This quadrilateral is a kite (two pairs of equal adjacent sides). A cyclic kite has two right angles, and it splits into two right triangles with legs 5 and 12 — each of area 30, giving 60 in total. The diagonal joining the two right angles is √(25+144) = 13, the diameter of the circumcircle.



AC = 13 is a diameter, so ∠B = ∠D = 90°; area = 2 × (½ × 5 × 12) = 60 sq units

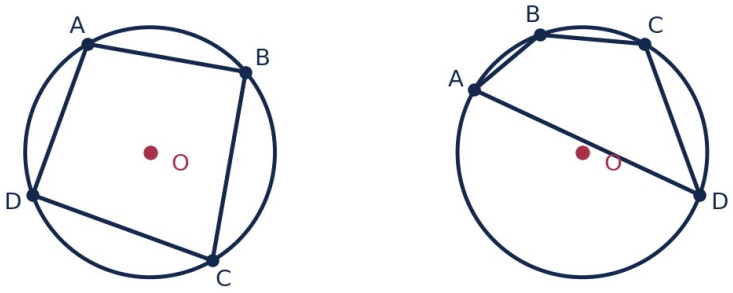
*11. How can we tell whether the circumcentre of a cyclic quadrilateral lies inside it, without drawing the circle?

Look at each side in turn and at the angle that side subtends at one of the two opposite vertices.

- If every side subtends an acute angle (less than 90°) at the opposite vertices, then every arc cut off by a side is less than a semicircle, and the centre lies inside the quadrilateral.
- If one side subtends a right angle, that side is a diameter and the centre lies exactly on that side.
- If one side subtends an obtuse angle (more than 90°), the arc it cuts off is more than a semicircle, and the centre lies outside the quadrilateral — on the far side of that particular side.

Reason: by Theorem 9, the arc cut off by a side equals twice the inscribed angle on it. An arc is less than a semicircle exactly when that inscribed angle is less than 90°.

Best practical method: construct the perpendicular bisectors of two of the sides; where they cross is the circumcentre, and you can see at a glance whether that point is inside or outside.



centre inside the quadrilateral *centre outside the quadrilateral*

When all four vertices are spread out the centre falls inside; when they crowd into one part of the circle it falls outside

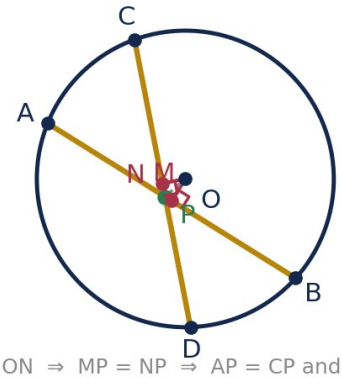
*12. Two equal chords intersect. Show that the segments of one equal the corresponding segments of the other.

Let equal chords AB and CD meet at P, with centre O. Let M and N be the midpoints of AB and CD.

Statement	Reason
OM ⊥ AB and ON ⊥ CD	line from centre to midpoint of a chord (Theorem 4)
OM = ON	equal chords are equidistant from the centre (Theorem 6)
In △OMP and △ONP: ∠OMP = ∠ONP = 90°	from the first line
OP = OP	common hypotenuse
OM = ON	proved above
∴ △OMP ≅ △ONP	by RHS congruence
∴ MP = NP	CPCT … (1)
AM = ½ AB = ½ CD = CN	M, N are midpoints of equal chords … (2)
AP = AM + MP and CP = CN + NP	adding along each chord

Statement	Reason
∴ AP = CP	adding (1) and (2)
Also BP = AB − AP = CD − CP = DP	subtracting from equal totals

So the two parts of one chord are equal to the two corresponding parts of the other.



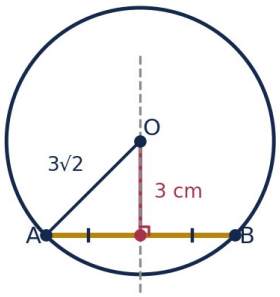
OM = ON ⇒ MP = NP ⇒ AP = CP and BP = DP

*13. Draw a circle in which a chord of length 6 cm is at a distance of 3 cm from the centre.

First find the radius: $r = \sqrt{(d^2 + (\text{half chord})^2)} = \sqrt{(3^2 + 3^2)} = \sqrt{18} = 3\sqrt{2} \approx 4.24$ cm.

Steps of construction: (1) Draw AB = 6 cm. (2) Construct the perpendicular bisector of AB, meeting it at M. (3) On this bisector mark O with OM = 3 cm. (4) With centre O and radius OA (= $3\sqrt{2} \approx 4.2$ cm), draw the circle. AB is then a chord of the required kind.

About the hint: yes — this circle is the circumcircle of the isosceles triangle OAB, and indeed of any triangle inscribed in it having AB as a side.



AB = 6 cm at distance 3 cm ⇒ r = 3√2 ≈ 4.24 cm

*14. Show that a rectangle is the only parallelogram that can be inscribed in a circle.

Let ABCD be a parallelogram inscribed in a circle.

Statement	Reason
∠A = ∠C	opposite angles of a parallelogram are equal
∠A + ∠C = 180°	opposite angles of a cyclic quadrilateral (Theorem 11)
∴ ∠A + ∠A = 180°, so 2∠A = 180°	substituting
∴ ∠A = 90°, and hence ∠C = 90°	dividing by 2
Similarly ∠B = ∠D = 90°	the same argument on the other pair
∴ all four angles are right angles, so ABCD is a rectangle	definition of a rectangle

*15. If a rectangle is inscribed in a circle, show that its diagonals meet at the centre.

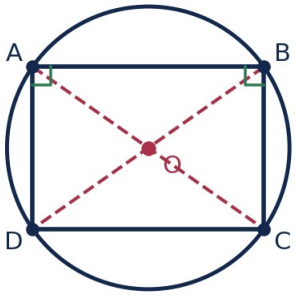
Let ABCD be the inscribed rectangle. Then ∠ABC = 90°.

So the diagonal AC subtends a right angle at the point B on the circle. By the converse of the angle-in-a-semicircle result, AC must be a diameter.

In the same way, since ∠BAD = 90°, the diagonal BD is also a diameter.

Every diameter passes through the centre, so both diagonals pass through the centre — therefore they meet there.

Alternatively: the diagonals of a rectangle bisect each other, so they meet at their common midpoint; and the midpoint of a diameter is the centre.



each diagonal is a diameter, so they meet at O

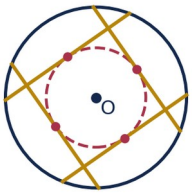
An inscribed rectangle: both diagonals are diameters

*16. What shape is formed by the midpoints of all chords of a fixed length?

A circle, concentric with the given circle (that is, having the same centre).

Reason: let the fixed chord length be L and the radius r. Every such chord is at distance $d = \sqrt{r^2 - L^2/4}$ from the centre (Theorem 6), and its midpoint is exactly the foot of the perpendicular from the centre.

So every one of these midpoints is at the same fixed distance d from the centre — and the set of points at a fixed distance from a fixed point is a circle of radius d.

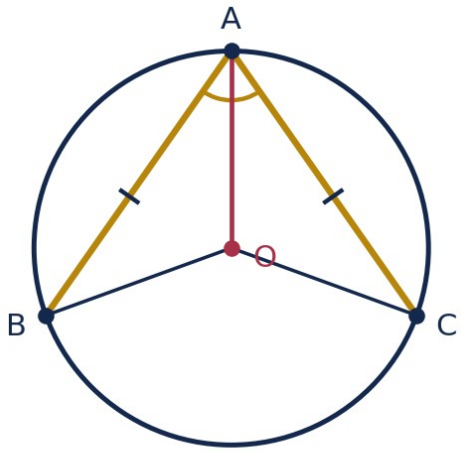


every midpoint is at the same distance d from O,
so the midpoints form a circle concentric with the given one

*17. Chords AB and AC are congruent. Explain why the centre lies on the bisector of ∠BAC.

Statement	Reason
In △OAB and △OAC: OB = OC	both are radii
AB = AC	given
OA = OA	common side
∴ △OAB ≅ △OAC	by SSS congruence
∴ ∠OAB = ∠OAC	CPCT
∴ AO bisects ∠BAC, so O lies on the bisector of ∠BAC	definition of an angle bisector

Another way to see it: equal chords are equidistant from the centre (Theorem 6), so O is equidistant from the two lines AB and AC — and the locus of points equidistant from two intersecting lines is the pair of angle bisectors.



AO bisects ∠BAC

18. Two parallel chords of lengths 10 cm and 24 cm lie on the same side of the centre, 7 cm apart. Find the radius.

Let the radius be r. Half the chords are 5 cm and 12 cm.

Distance of the 10 cm chord from the centre: $d_1 = \sqrt{(r^2 - 25)}$.

Distance of the 24 cm chord from the centre: $d_2 = \sqrt{(r^2 - 144)}$.

The 24 cm chord is longer, so it is nearer the centre. As both are on the same side, the distances subtract:

$d_1 - d_2 = 7$, that is $\sqrt{(r^2 - 25)} - \sqrt{(r^2 - 144)} = 7$ … (1)

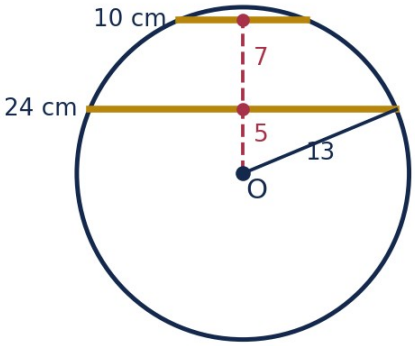
Also $d_1^2 - d_2^2 = (r^2 - 25) - (r^2 - 144) = 119$.

Since $d_1^2 - d_2^2 = (d_1 - d_2)(d_1 + d_2)$, we get $7\,(d_1 + d_2) = 119$, so $d_1 + d_2 = 17$ … (2)

Adding (1) and (2): $2d_1 = 24$, so $d_1 = 12$. Then $d_2 = 5$.

Finally $r^2 = d_1^2 + 25 = 144 + 25 = 169$.

Answer: r = 13 cm.



same side: 12 − 5 = 7 cm ⇒ r = 13 cm

***19.** A regular hexagon is inscribed in a circle of radius r. Find its side and the distance of each side from the centre.

Join the centre O to two adjacent vertices A and B. The six equal central angles make up 360°, so ∠AOB = 360° ÷ 6 = 60°.

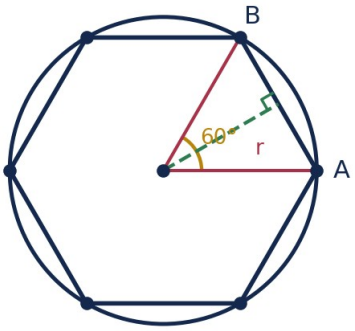
In △OAB, OA = OB = r and the included angle is 60°, so the base angles are each (180° − 60°) ÷ 2 = 60°. The triangle is therefore equilateral.

Side of the hexagon: AB = r.

The distance of the side from the centre is the perpendicular from O to AB, which bisects AB, giving a right triangle with hypotenuse r and one leg r/2:

distance = $\sqrt{(r^2 - (r/2)^2)} = \sqrt{(3r^2/4)}$

Distance of each side from the centre: $(\sqrt{3}/2)\,r \approx 0.87\,r$.



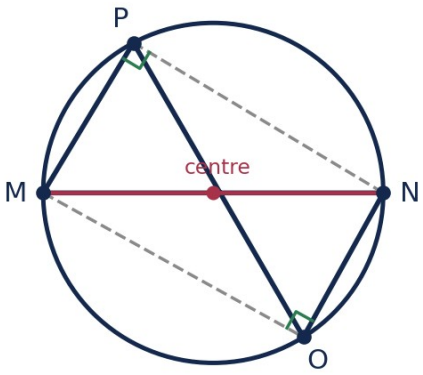
side AB = r, distance from centre = (√3/2) r

20. Quadrilateral MNOP is inscribed in a circle with MN a diameter. What can you say about ∠MOP and ∠MNP?

They are equal: ∠MOP = ∠MNP.

Reason: both angles stand on the same chord MP, and the vertices O and N lie on the same arc (the one on the same side of MP). By Theorem 9, angles in the same segment are equal.

Using the diameter as well: since MN is a diameter, the angle it subtends at any point of the circle is 90°. So ∠MON = 90° and ∠MPN = 90° — the angles at O and at P standing on MN are both right angles.



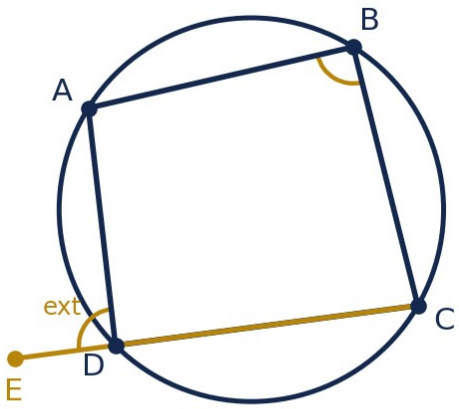
MN is a diameter ⇒ ∠MON = ∠MPN = 90°

21. Show that the exterior angle of a cyclic quadrilateral equals the interior opposite angle.

Let ABCD be cyclic and let E lie on CD produced, so that ∠CDE is the exterior angle at D.

Statement	Reason
∠CDE + ∠ADC = 180°	linear pair, since CDE is a straight line
∠ABC + ∠ADC = 180°	opposite angles of a cyclic quadrilateral (Theorem 11)
∴ ∠CDE + ∠ADC = ∠ABC + ∠ADC	both equal 180°
∴ ∠CDE = ∠ABC	cancelling ∠ADC from both sides

So the exterior angle at any vertex equals the interior angle at the opposite vertex.

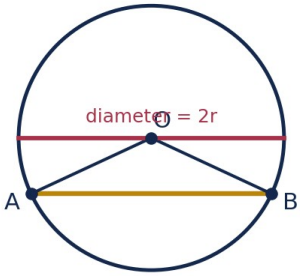


∠CDE = ∠ABC

*22. Justify: "There is no chord of a circle longer than its diameter."

Method 1 (formula): a chord at distance d from the centre has length $2\sqrt{(r^2 - d^2)}$. Since $d^2 \geq 0$, we get $r^2 - d^2 \leq r^2$, so the chord length is at most 2r — the diameter. Equality happens only when d = 0, i.e. when the chord passes through the centre.

Method 2 (triangle inequality): for any chord AB, join OA and OB. If O is not on AB then AB < OA + OB = 2r. If O is on AB then AB = 2r exactly. So no chord exceeds 2r.



AB < OA + OB = 2r, unless AB passes through O

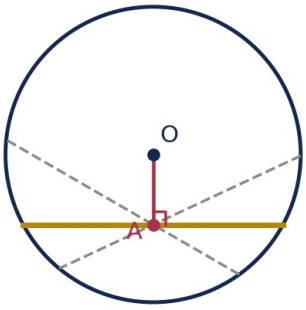
*23. A is a point inside a circle with centre O. Show that the shortest chord through A is the one perpendicular to OA.

Any chord through A has length $2\sqrt{(r^2 - d^2)}$, where d is the distance of that chord from O. This length is smallest when d is largest.

Let a chord through A have foot of perpendicular M from O. Then △OMA is right-angled at M (or M coincides with A), so OM ≤ OA, with equality only when M and A are the same point.

So the greatest possible value of d is OA itself, achieved exactly when the foot of the perpendicular is A — that is, when OA ⊥ the chord.

Conclusion: the shortest chord through A is the one perpendicular to OA, and its length is $2\sqrt{(r^2 - OA^2)}$.



the shortest chord through A is the one ⊥ to OA

24. Use Fig. 5.30 to justify that the angle in a semicircle is 90°.

In the figure, BC is a diameter, O is the centre and A is a point on the circle. Join OA.

Statement	Reason
OA = OB, both radii, so △OAB is isosceles	radii of the same circle
∴ ∠OAB = ∠OBA = a	base angles of an isosceles triangle
OA = OC, both radii, so △OAC is isosceles	radii of the same circle
∴ ∠OAC = ∠OCA = b	base angles of an isosceles triangle
In △ABC: ∠BAC + ∠ABC + ∠ACB = 180°	angle sum of a triangle
∠BAC = ∠OAB + ∠OAC = a + b	the two parts make up the angle at A
∴ (a + b) + a + b = 180°	substituting
∴ 2(a + b) = 180°, so a + b = 90°	dividing by 2
∴ ∠BAC = a + b = 90°	the angle in a semicircle

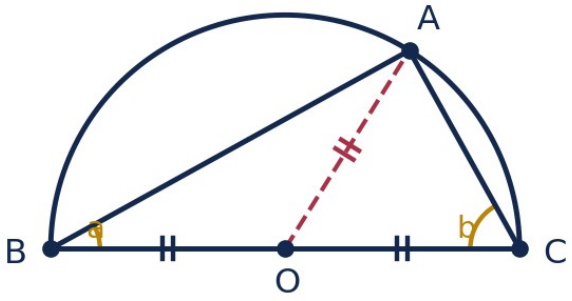


Fig. 5.30 — joining OA creates two isosceles triangles

***25.** Chords CC′ and DD′ are perpendicular to a diameter AB. Prove that MM′ (joining the midpoints of CD and C′D′) is perpendicular to AB.

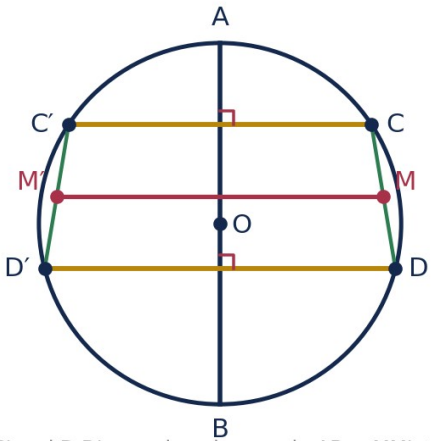
The diameter AB passes through the centre O, and CC′ ⊥ AB. By Theorem 5, the perpendicular from the centre to the chord CC′ bisects it — so AB bisects CC′ at right angles.

That means C and C′ are mirror images of each other in the line AB. For the same reason, D and D′ are mirror images in AB.

Reflection in a line preserves lengths and midpoints. So the segment C′D′ is the mirror image of CD, and therefore M′ (the midpoint of C′D′) is the mirror image of M (the midpoint of CD).

Finally, the segment joining any point to its mirror image is always perpendicular to the mirror line.

Therefore MM′ ⊥ AB.



C,C′ and D,D′ are mirror images in AB ⇒ MM′ ⊥ AB

***26.** Use Fig. 5.31 to justify that opposite angles of a cyclic quadrilateral add to 180°.

In the figure, O is the centre of the circle and OA, OB, OC, OD are joined. This splits the quadrilateral into four triangles, and each is isosceles because two of its sides are radii.

Let the base angles be: p at A and at B in △OAB; q at B and at C in △OBC; u at C and at D in △OCD; v at D and at A in △ODA.

The four angles of quadrilateral ABCD are made from these pieces:

$\angle A = p + v$, $\angle B = p + q$, $\angle C = q + u$, $\angle D = u + v$

Adding all four: $\angle A + \angle B + \angle C + \angle D = 2(p + q + u + v)$.

But the angle sum of any quadrilateral is 360°, so $2(p + q + u + v) = 360°$, giving $p + q + u + v = 180°$.

Now $\angle A + \angle C = (p + v) + (q + u) = p + q + u + v = 180°$.

Similarly $\angle B + \angle D = (p + q) + (u + v) = 180°$. So the opposite angles are supplementary.

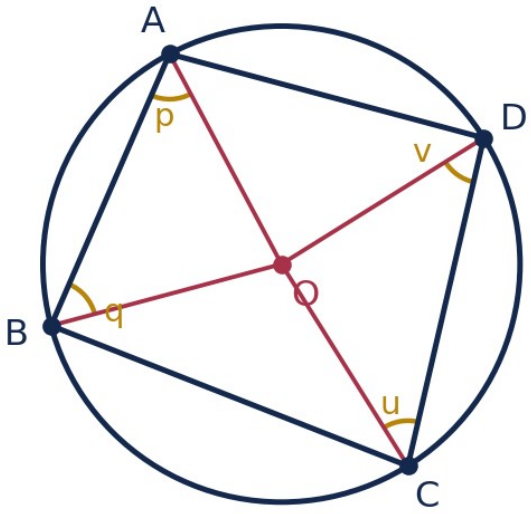


Fig. 5.31 — the centre joined to all four vertices makes four isosceles triangles

PART C — ONE-PAGE REVISION SHEET

Definitions

- Circle: all points of a plane at a fixed distance (the radius) from a fixed point (the centre).
- Chord: a segment joining two points of the circle. Diameter: a chord through the centre; it is the longest chord and equals 2r.
- Arc: a piece of the circle. Two points give a minor arc and a major arc.
- Concyclic points: points lying on one circle. Cyclic quadrilateral: a quadrilateral whose vertices are concyclic.
- Circumcircle: the circle through the three vertices of a triangle; its centre is the circumcentre.

The twelve theorems at a glance

- T1 Exactly one circle passes through three non-collinear points.
- T2 Equal chords subtend equal angles at the centre. (SSS)
- T3 Equal angles at the centre give equal chords. (SAS) — converse of T2
- T4 Centre to the midpoint of a chord is perpendicular to it. (SSS)
- T5 The perpendicular from the centre bisects the chord. (RHS) — converse of T4
- T6 Equal chords are equidistant from the centre. (RHS)
- T7 Chords equidistant from the centre are equal. (RHS) — converse of T6
- T8 Of two unequal chords, the longer one is nearer the centre.
- T9 Angle at the centre = 2 × angle at any point of the circle on the remaining part.
- Corollary The angle in a semicircle is 90°.
- T10 Equal angles on the same side of a segment ⇒ the four points are concyclic.
- T11 Opposite angles of a cyclic quadrilateral add to 180°.
- T12 If opposite angles add to 180°, the quadrilateral is cyclic. — converse of T11

Formulas you must know

Chord length = $2\sqrt{(r^2 - d^2)}$ (d = perpendicular distance of the chord from the centre)

Distance from centre = $\sqrt{(r^2 - (\tfrac{1}{2} \times \text{chord})^2)}$

$r^2 = d^2 + (\tfrac{1}{2} \times \text{chord})^2$ — the master right-triangle relation

Brahmagupta (cyclic quadrilateral): Area = $\sqrt{(s-a)(s-b)(s-c)(s-d)}$

Common traps in the exam

- Parallel chords on OPPOSITE sides of the centre: add the two distances. On the SAME side: subtract them.
- Chord length is not proportional to distance — always go back to $r^2 = d^2 + (\tfrac{1}{2}\text{ chord})^2$.
- Angles in the same segment are equal only when the points are on the same side of the chord.
- Three collinear points have no circle through them — the perpendicular bisectors are parallel.
- In any proof, write Given, To prove, Construction (if any), then the Proof with a reason for every step.